

title: "Premier League 19-20 Data Analysis & Visualization" author: "Eric Turba" date: "2026-01-15" output: pdf_document —

```
library(tidyverse)

## Warning: package 'lubridate' was built under R version 4.5.2

## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4    ✓ readr      2.1.5
## ✓ forcats    1.0.1    ✓ stringr    1.5.2
## ✓ ggplot2     4.0.0    ✓ tibble     3.3.0
## ✓ lubridate  1.9.4    ✓ tidyr      1.3.1
## ✓ purrr      1.1.0
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(readxl)
library(GGally)
library(gridExtra)

##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##   combine

library(DT)
library(ggimage)
premierLeague <- read_excel("premierLeague.xlsx")
datatable(premierLeague)
```

Show entries Search:

	Match_Name	Match_Date	Team	Team_Logo	Manager	Cap
1	Liverpool vs. Norwich City	2019-08-09T00:00:00Z	Liverpool	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/822bd0ba.png	Jürgen Klopp	Jord Hen
2	Liverpool vs. Norwich City	2019-08-09T00:00:00Z	Norwich City	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/1c781004.png	Daniel Farke	Gran Han
3	West Ham United vs. Manchester City	2019-08-10T00:00:00Z	West Ham United	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/7c21e445.png	Manuel Pellegrini	Aarc Cres
4	West Ham United vs. Manchester City	2019-08-10T00:00:00Z	Manchester City	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/b8fd03ef.png	Pep Guardiola	Davi Silve
5	Burnley vs. Southampton	2019-08-10T00:00:00Z	Burnley	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/943e8050.png	Sean Dyche	Ben
6	Burnley vs. Southampton	2019-08-10T00:00:00Z	Southampton	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/33c895d4.png	Ralph Hasenhüttl	Jam War Prov
7	Watford vs. Brighton & Hove Albion	2019-08-10T00:00:00Z	Watford	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/2abfe087.png	Javi Gracia	Troy Dee
8	Watford vs. Brighton & Hove Albion	2019-08-10T00:00:00Z	Brighton & Hove Albion	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/d07537b9.png	Graham Potter	Lew Dun
9	Bournemouth vs. Sheffield United	2019-08-10T00:00:00Z	Bournemouth	https://d2p3bygnnzw9w3.cloudfront.net/req/202007313/tlogo/fb/4ba7cbea.png	Eddie Howe	Stev Coo

	Match_Name	Match_Date	Team	Team_Logo	Manager	Cap
10	Bournemouth vs. Sheffield United	2019-08- 10T00:00:00Z	Sheffield United	https://d2p3byggnzw9w3.cloudfront.net/req/202007313/tlogo/fb/1df6b87e.png	Chris Wilder	Oliv Non

Showing 1 to 10 of 760 entries

Previous

1

2

3

4

5

...

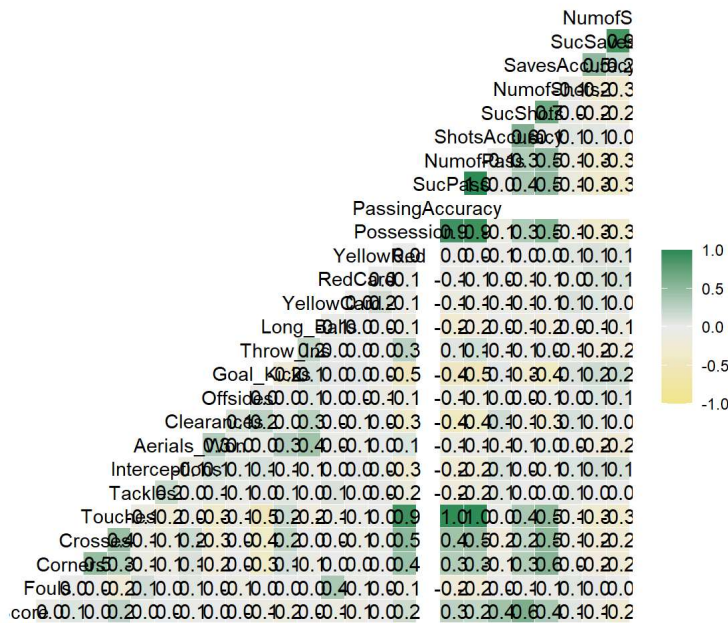
76

Next

Correlation

```
ggcorr(premierLeague %>%
  select_if(is.numeric) %>%
  select(-Penalties),label = TRUE, digits = 1,low = "khaki", high = "seagreen")
```

```
## Warning in cor(data, use = method[1], method = method[2]): the standard
## deviation is zero
```

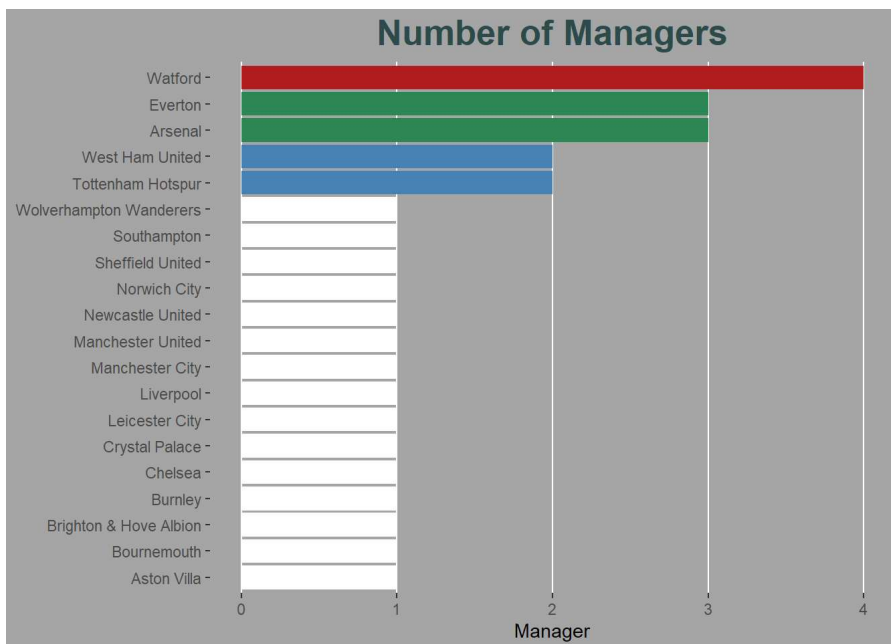


```

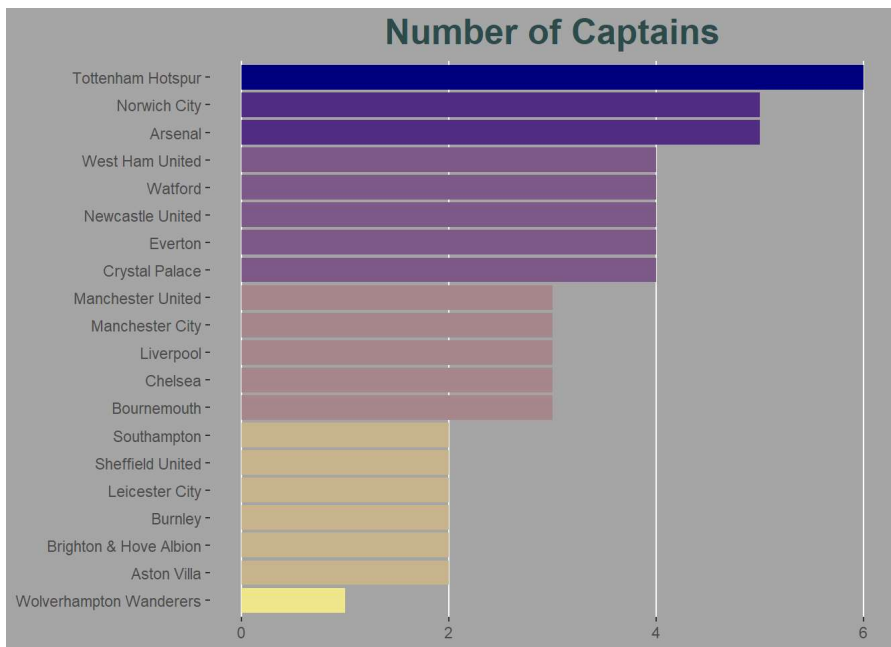
# Number of Managers & Captains
p1 <- premierLeague %>%
  group_by(Team) %>%
  summarise(Manager = length(unique(Manager))) %>%
  ungroup() %>%
  arrange(-Manager) %>%
  mutate(Cat = case_when(
    Manager == 1 ~ "1",
    Manager == 2 ~ "2",
    Manager == 3 ~ "3",
    Manager > 3 ~ "4+"
  )) %>%
  ggplot(aes(reorder(Team, Manager), Manager, fill = Cat))+
  geom_col()+
  coord_flip()+
  labs(x = NULL, x = NULL, fill = "Number of Managers", title = "Number of Managers")+
  theme(
    panel.background = element_rect(fill = "darkgray"),
    plot.background = element_rect(fill = "darkgray", color = "darkgray"),
    legend.background = element_rect(fill = "darkgray"),
    legend.key = element_blank(),
    panel.grid.major.y = element_blank(),
    panel.grid.minor.x = element_blank(),
    legend.position = "bottom",
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5, color = "darkslategray")
  )+
  scale_fill_manual(values = c("white", "steelblue", "seagreen", "firebrick"))

p2 <- premierLeague %>%
  group_by(Team) %>%
  summarize(Captain = length(unique(Captain))) %>%
  ungroup() %>%
  arrange(-Captain) %>%
  ggplot(aes(reorder(Team, Captain), Captain, fill = Captain))+
  geom_col(show.legend = FALSE)+
  coord_flip()+
  labs(x = NULL, y = NULL, title = "Number of Captains")+
  scale_fill_gradient(low = "khaki", high = "navy")+
  theme(
    panel.background = element_rect(fill = "darkgray"),
    plot.background = element_rect(fill = "darkgray", color = "darkgray"),
    legend.background = element_rect(fill = "darkgray"),
    legend.key = element_blank(),
    panel.grid.major.y = element_blank(),
    panel.grid.minor.x = element_blank(),
    legend.position = "bottom",
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5, color = "darkslategray")
  )
)
p1

```



p2



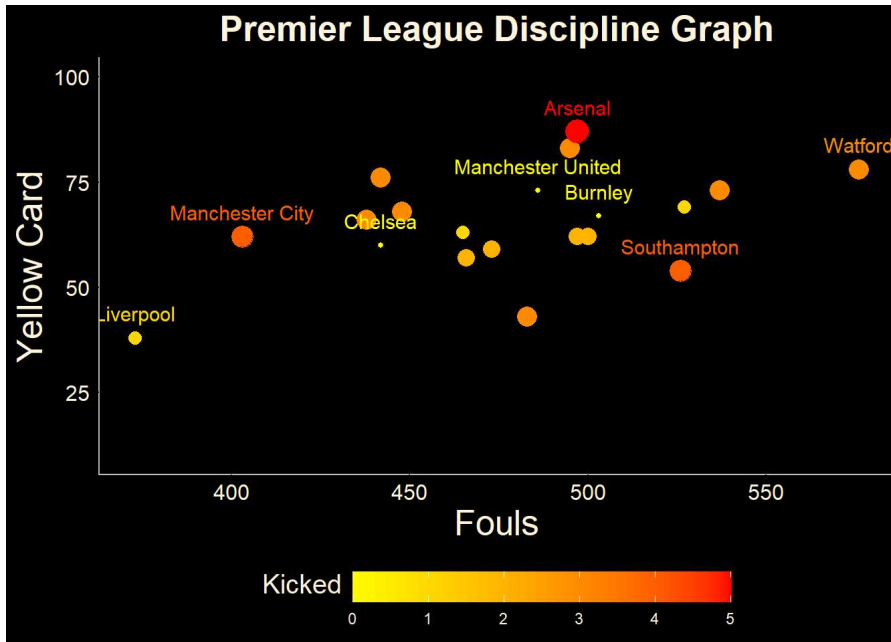
```
# Discipline
discipline <- premierLeague %>%
  select(Team, Fouls, YellowCard, RedCard, YellowRed) %>%
  group_by(Team) %>%
  summarise_at(vars(Fouls, YellowCard:YellowRed), funs(sum)) %>%
  arrange(Fouls) %>%
  mutate(Disqualified = RedCard+YellowRed)
```

```
## Warning: `funs()` was deprecated in dplyr 0.8.0.
## i Please use a list of either functions or lambdas:
##
## # Simple named list: list(mean = mean, median = median)
##
## # Auto named with `tibble::lst()` : tibble::lst(mean, median)
##
## # Using lambdas list(~ mean(. , trim = .2), ~ median(. , na.rm = TRUE))
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
disp2 <- discipline %>%
  arrange(Disqualified) %>%
  filter(Team %in% c("Chelsea", "Manchester United", "Burnley", "Liverpool", "Arsenal", "Southampton", "Manchester City", "W
atford"))

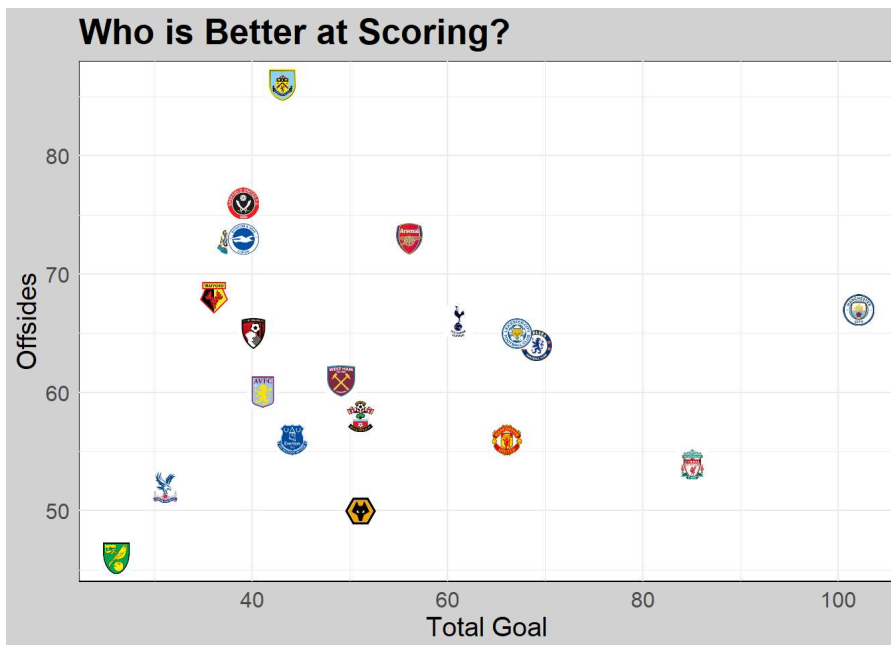
ggplot(discipline, aes(x = Fouls, YellowCard, color = Disqualified, size = Disqualified))+
  geom_point()+
  geom_text(disp2, mapping = aes(Fouls, YellowCard, label = Team), size = 4, vjust = -1.1)+
  guides(size=FALSE)+
  scale_color_gradient(high = "red", "low" = "yellow")+
  theme_classic()+
  theme(
    # Background
    plot.background = element_rect(fill = "black", color = "black"),
    panel.background = element_rect(fill = "black"),
    legend.background = element_rect(fill = "black"),
    # Position
    legend.position = "bottom",
    legend.key.width=unit(1.5,"cm"),
    # Text
    axis.title = element_text(color = "cornsilk", size = 20),
    axis.text = element_text(color = "cornsilk", size = 12),
    legend.text = element_text(color = "cornsilk"),
    legend.title = element_text(color = "cornsilk", vjust = 0.95, size = 15),
    # Line
    axis.line = element_line(color = "gray"),
    # Title
    plot.title = element_text(hjust = .5, color = "cornsilk", size = 20, face = "bold")
  )+
  ylim(10,100)+
  labs(y = "Yellow Card", color = "Kicked", title = "Premier League Discipline Graph")
```

```
## Warning: The `<scale>` argument of `guides()` cannot be `FALSE`. Use "none" instead as
## of ggplot2 3.3.4.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



```
# Who is better at scoring
premierLeague %>%
  select(Team_Logo, Team, Score, Offsides) %>%
  group_by(Team_Logo, Team) %>%
  summarise_at(vars(Score, Offsides), funs(sum)) %>%
  arrange(Offsides) %>%
  ggplot(aes(Offsides, Score))+
  geom_image(aes(image = Team_Logo), size = 0.06)+
  coord_flip()+
  theme_minimal()+
  theme(
    # Background
    plot.background = element_rect(fill = "lightgray"),
    panel.background = element_rect(fill = "white"),
    # Title
    plot.title = element_text(size = 20, face = "bold"),
    # Text
    axis.title = element_text(size = 15),
    axis.text = element_text(size = 12)
  )+
  labs(y = "Total Goal", title = "Who is Better at Scoring?")
```

```
## Warning: `funs()` was deprecated in dplyr 0.8.0.
## i Please use a list of either functions or lambdas:
##
## # Simple named list: list(mean = mean, median = median)
##
## # Auto named with `tibble::lst()`: tibble::lst(mean, median)
##
## # Using lambdas list(~ mean(. , trim = .2), ~ median(. , na.rm = TRUE))
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



```
# Who are successful managers
teams <- premierLeague %>%
  group_by(Team) %>%
  summarise(L = length(unique(Manager))) %>%
  ungroup() %>%
  filter(L == 1) %>% pull(Team)

premierLeague %>%
  filter(Team %in% teams) %>%
  group_by(Manager, Team) %>%
  summarise(Score = sum(Score), ShotsAccuracy = mean(ShotsAccuracy)) %>%
  ungroup() %>%
  ggplot(aes(Score, ShotsAccuracy, label = paste0(Manager, " (", Team, ")")))+
  geom_rect(aes(xmin = 20, xmax = 52.53, ymin = 0.275, ymax = 0.3322), fill = "orangered")+
  geom_rect(aes(xmin = 52.53, xmax = 110, ymin = 0.3322, ymax = 0.4), fill = "seagreen")+
  geom_hline(aes(yintercept = 0.3322), color = "cornsilk")+
  geom_vline(aes(xintercept = 52.53), color = "cornsilk")+
  geom_point(aes(x = 52.53, y = 0.3322), size = 4, color = "khaki")+
  geom_text(aes(x = 56, y = 0.335, label = "Season Average"), color = "khaki")+
  geom_text(vjust = -0.6, color = "white", size = 3)+
  geom_jitter(alpha = 0.4, color = "white")+
  xlim(20, 110)+ylim(0.275,0.4)+
  labs(y = "Average Shot Accuracy", title = "Who are successful managers?")+
  scale_y_continuous(labels = scales::percent)+
  theme_classic()+
  theme(
    # Background
    plot.background = element_rect(fill = "black", color = "black"),
    panel.background = element_rect(fill = "black"),
    legend.background = element_rect(fill = "black"),
    # Position
    legend.position = "bottom",
    legend.key.width = unit(1.5,"cm"),
    # Text
    axis.title = element_text(color = "cornsilk", size = 20),
    axis.text = element_text(color = "cornsilk", size = 12),
    legend.text = element_text(color = "cornsilk"),
    legend.title = element_text(color = "cornsilk", vjust = 0.95, size = 15),
    # Line
    axis.line = element_line(color = "gray"),
    # Title
    plot.title = element_text(hjust = .5, color = "cornsilk", size = 20, face = "bold")
  )
```

```
## `summarise()` has grouped output by 'Manager'. You can override using the
## `.groups` argument.
## Scale for y is already present. Adding another scale for y, which will replace
## the existing scale.
```

```
## Warning in geom_text(aes(x = 56, y = 0.335, label = "Season Average"), color = "khaki"): All aesthetics have length 1, but the data has 15 rows.  
## i Please consider using `annotate()` or provide this layer with data containing  
## a single row.
```

